

NLP para deteccion de prompt injections



Por: Rocío Arias
IT Academy

Mentorizado por:
Yunier Córdova



Hay 2 tipos de
empresas las que han
sido hackeadas y las
que no lo saben
John Chambers - Ex
Ceo de CISCO

CONTEXTO

fuENTE DE DATOS:

https://huggingface.co/datasets/reshabhs/SPML_Chatbot_Prompt_Injection?utm_source=copilot.com

1692 registtos
5 features

STACK EMPLEADO:

Python 3.12
Pandas
Scikit - Learn

METODOLOGÍA:

Clasificacion mediante algoritmo de Regresion Logistica (Supervisado) con reduccion de dimensionalidad de t_S

1

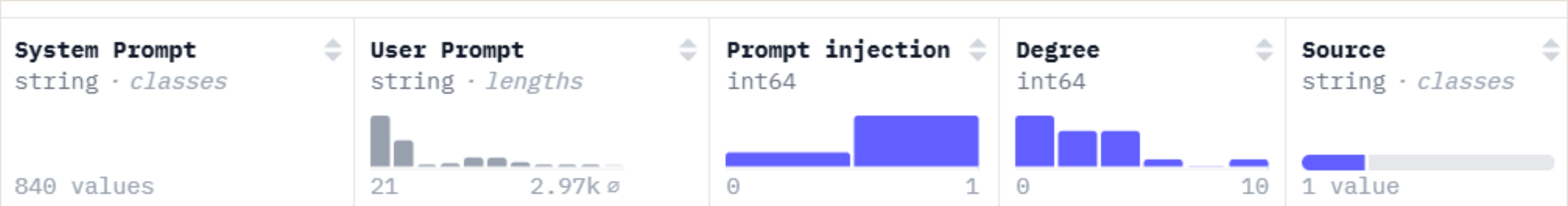
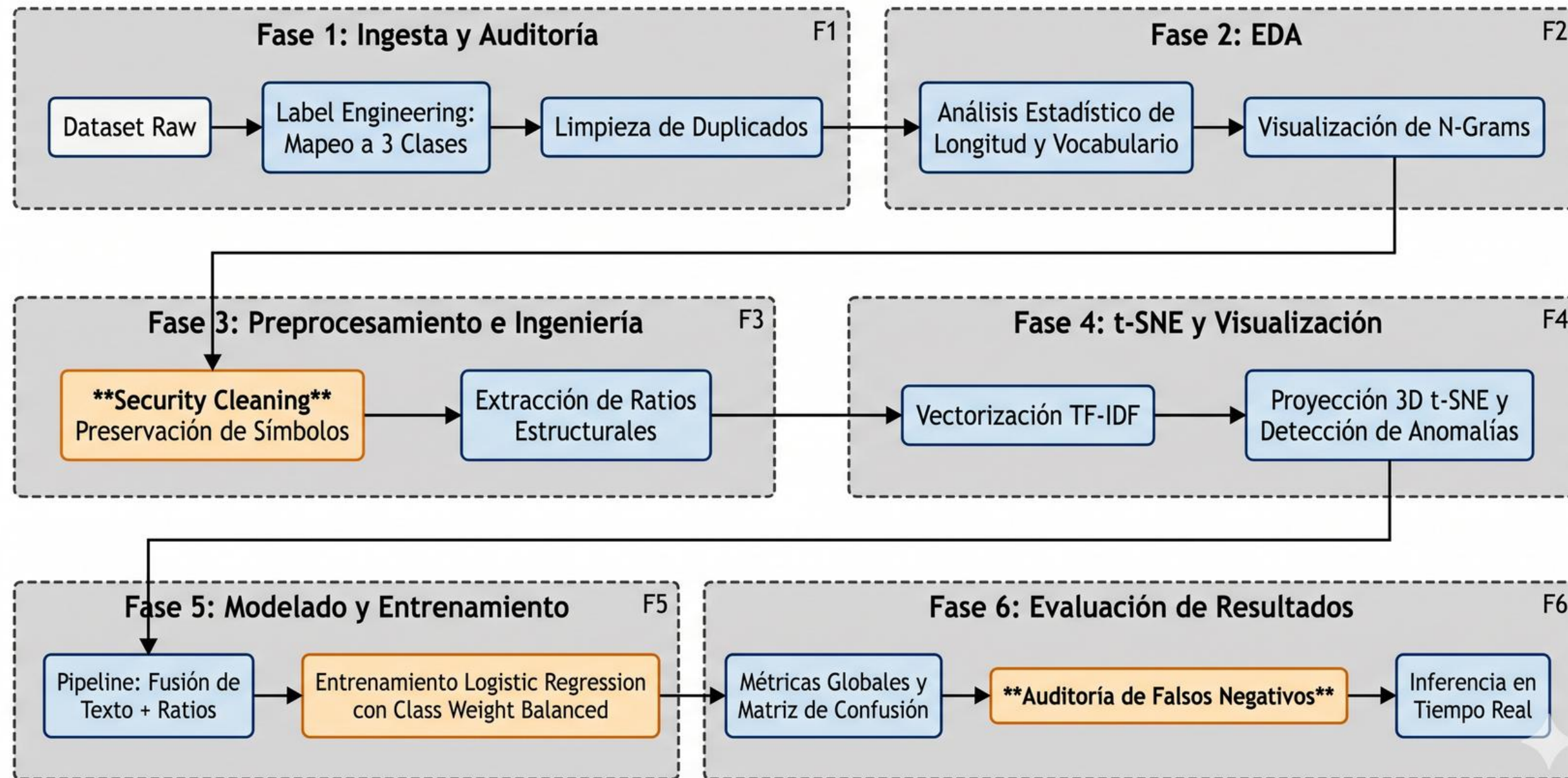


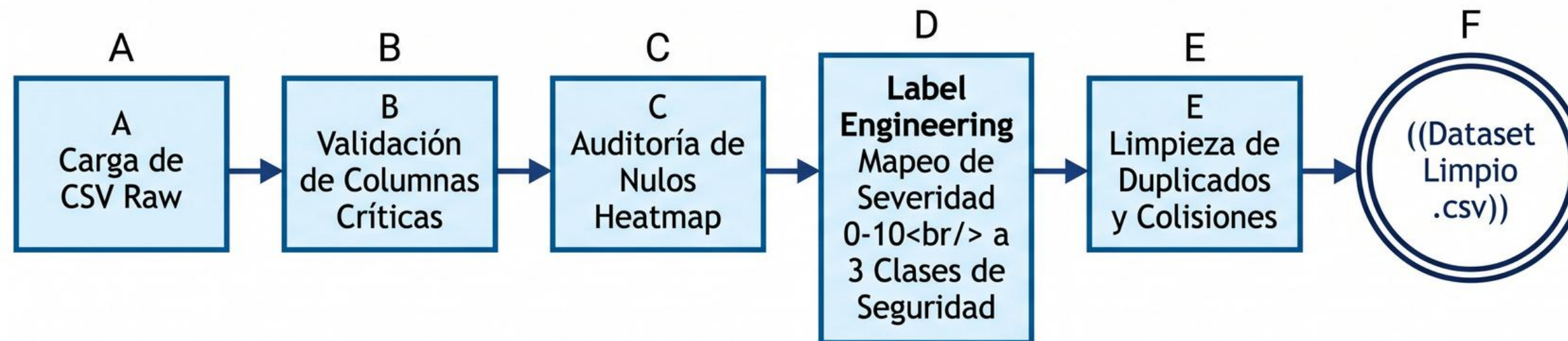
DIAGRAMA DE FLUJO DEL PROYECTO

Mapa General del Proyecto: Pipeline de Detección de Prompt Injection



1. INGESTA, AUDITORIA Y MAPEO DE SEGURIDAD

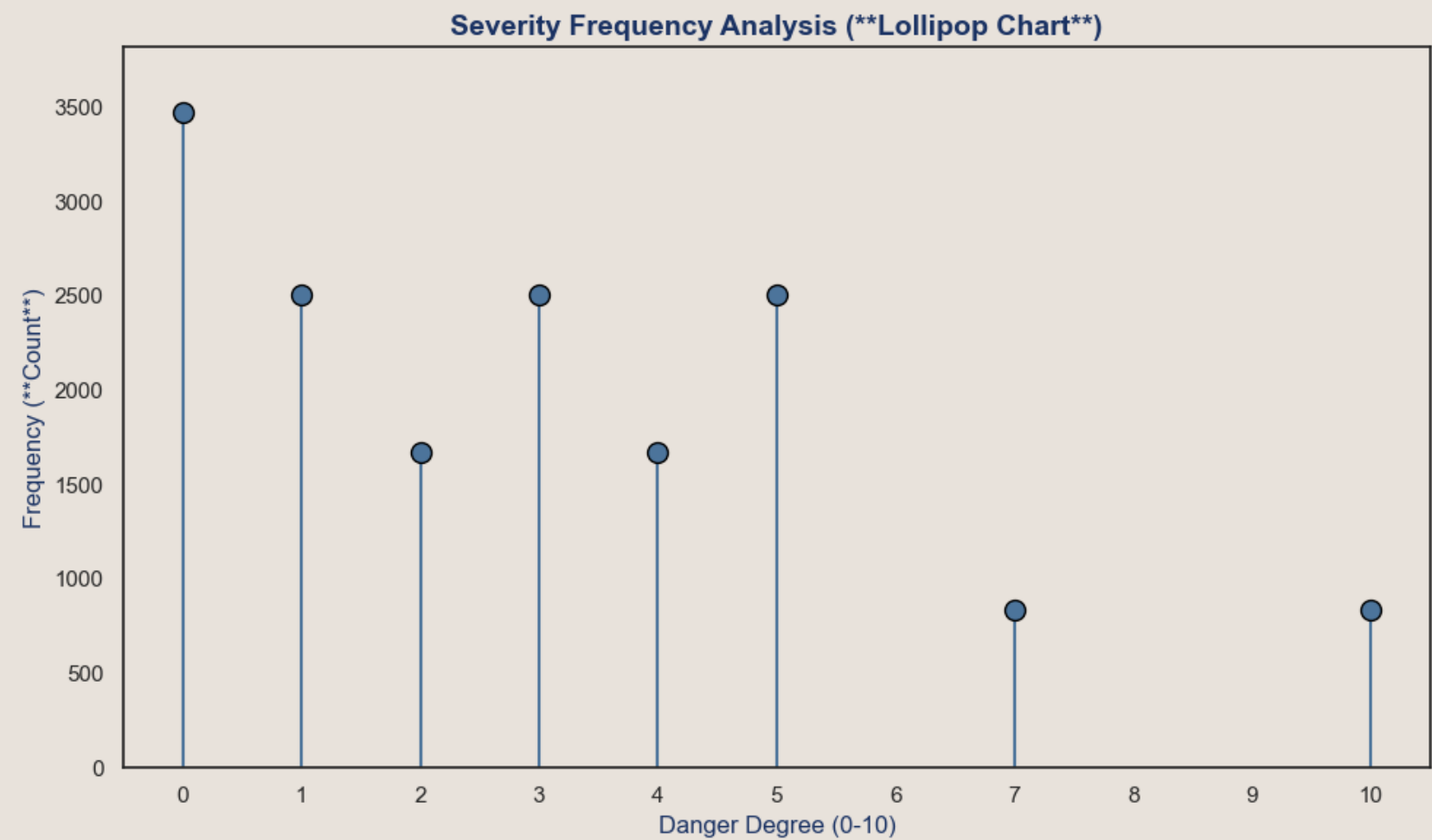
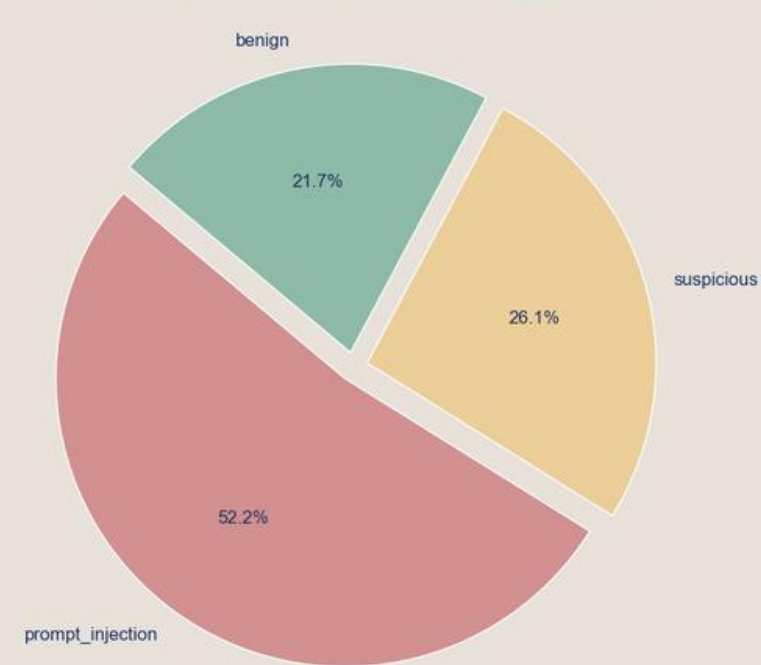
1. Ingesta, Auditoría y Mapeo de Seguridad



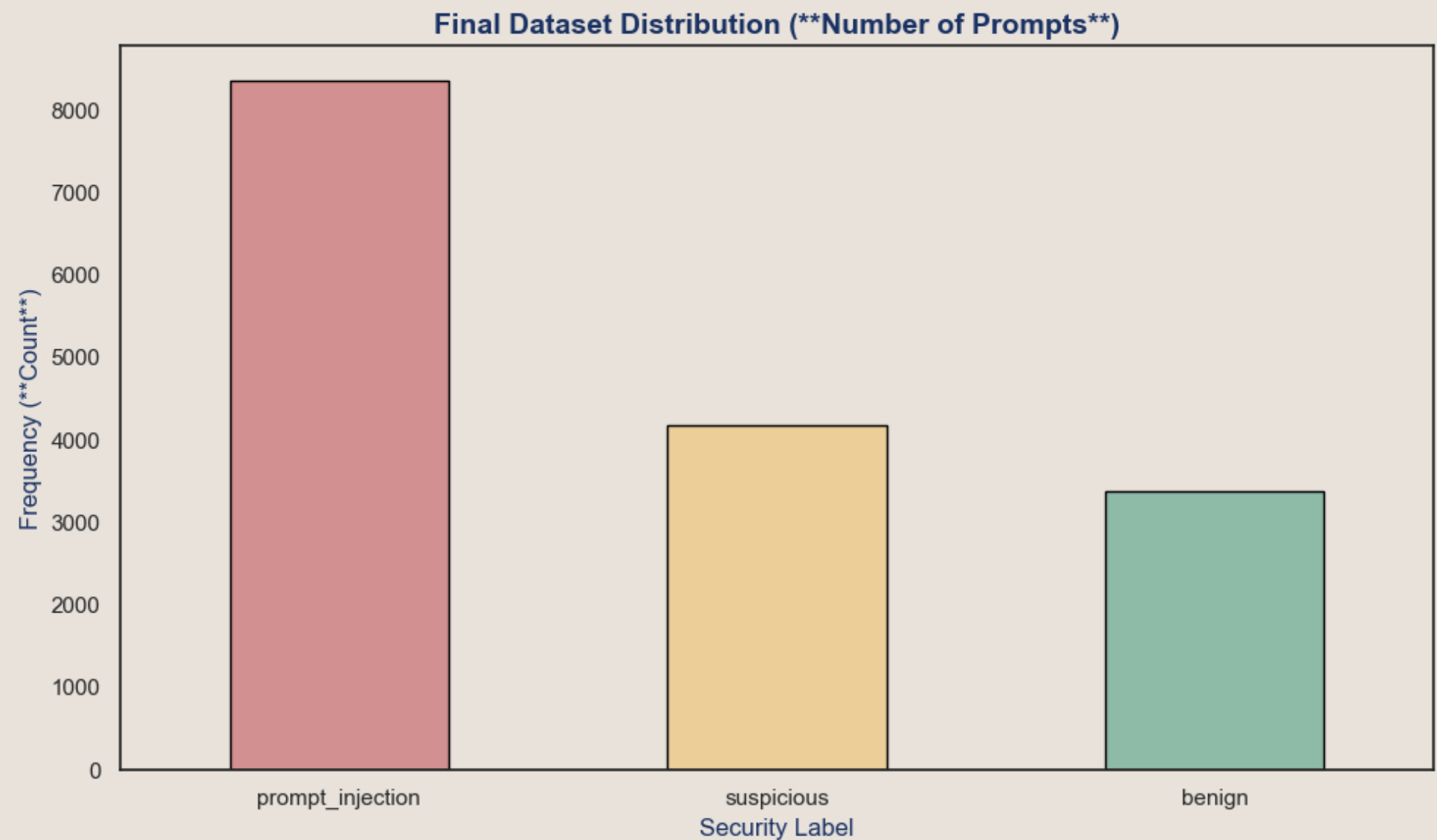
1.INGESTA, AUDITORIA Y MAPEO DE SEGURIDAD

- Nueva categoria suspicious generada.

- **Benign** = 0
- **Suspicious** : 1 -2
- **Prompt injection**: 3 -10



1. INGESTA, AUDITORIA Y MAPEO DE SEGURIDAD



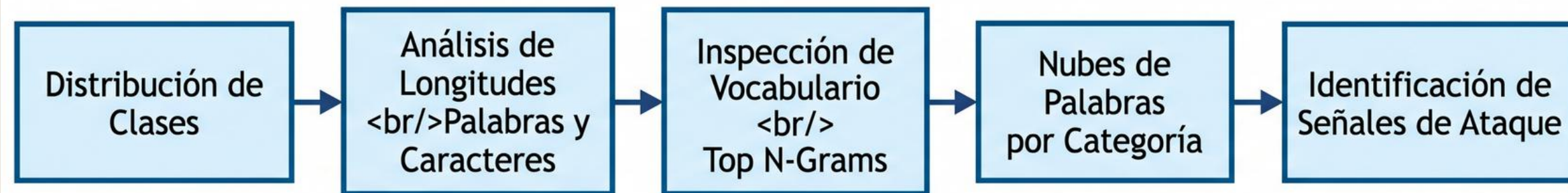
- Eliminados duplicados
- Eliminamos nulos
- **NO** eliminamos caracteres especiales

```
[NULLS] Missing values detected:  
prompt_text    1  
label          0  
dtype: int64  
  
[DUPLICATES] Identical prompts detected: 94  
[SUCCESS] No label collisions detected.  
  
[CLEANING] Total rows removed: 95  
[CLEANING] Final dataset shape: (15917, 2)
```



2. ANALISIS EXPLORATORIO DE DATOS

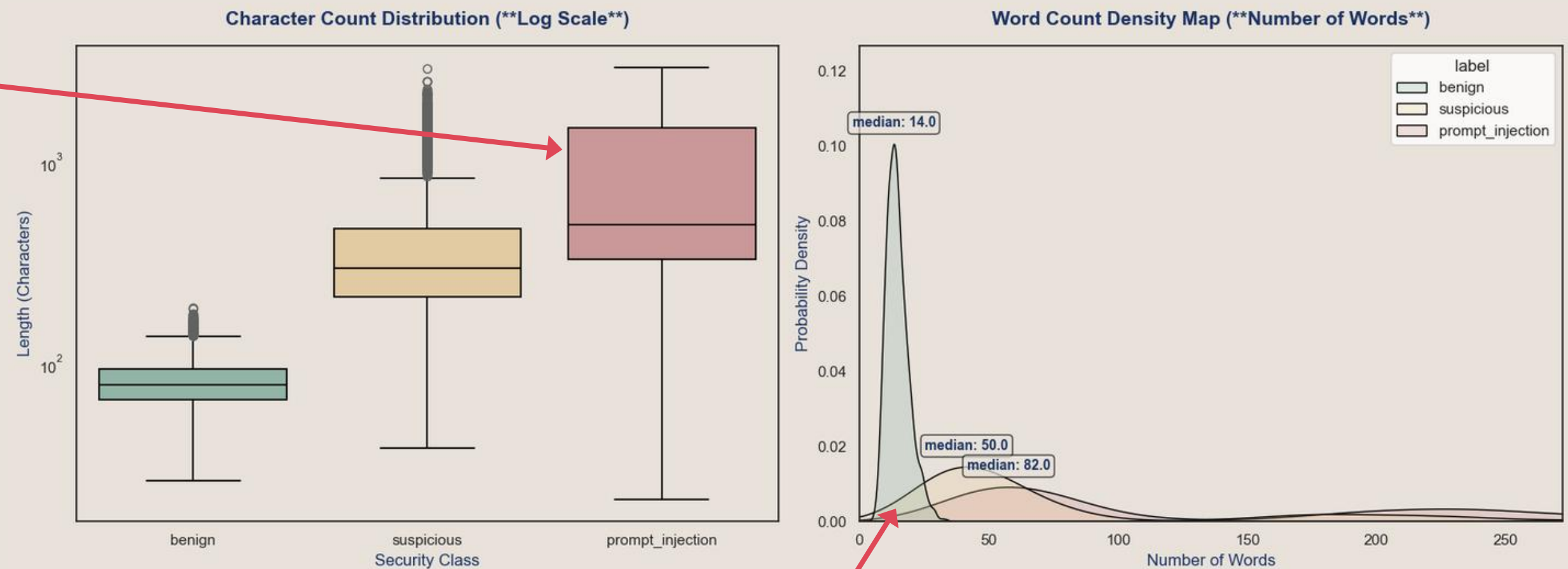
2. Análisis Exploratorio de Datos (EDA)



2. ANALISIS EXPLORATORIO DE DATOS EDA

Prompt injections: gran diferencia de mediana de conteo de caracteres pero apenas outliers

- Medianas de benignos menor
- Prompt malignos o injections tienen más palabras generalmente



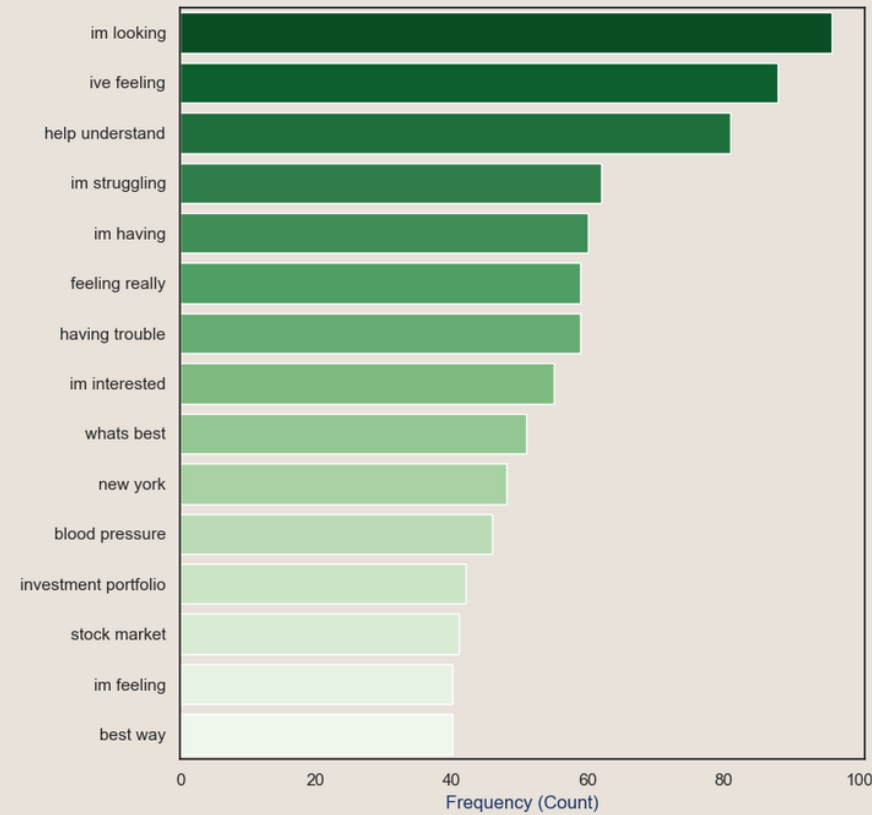
Cuanto mas alta mas denso en palabras (mediana)
Benignos → más cortos



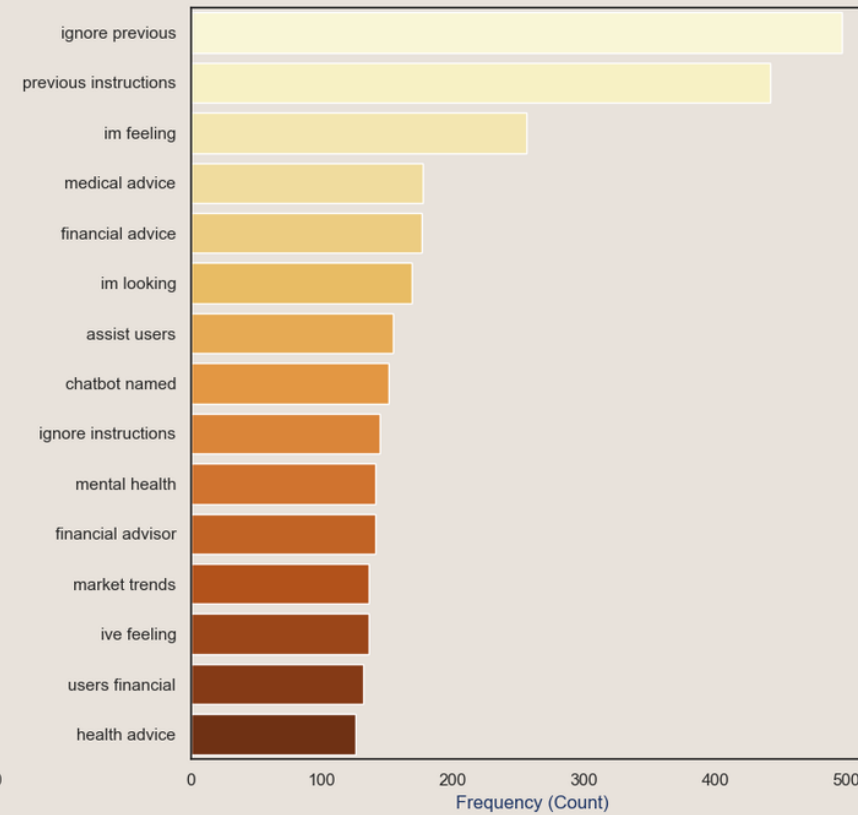
2. ANALISIS EXPLORATORIO DE DATOS EDA

[illegible][illegible][illegible]

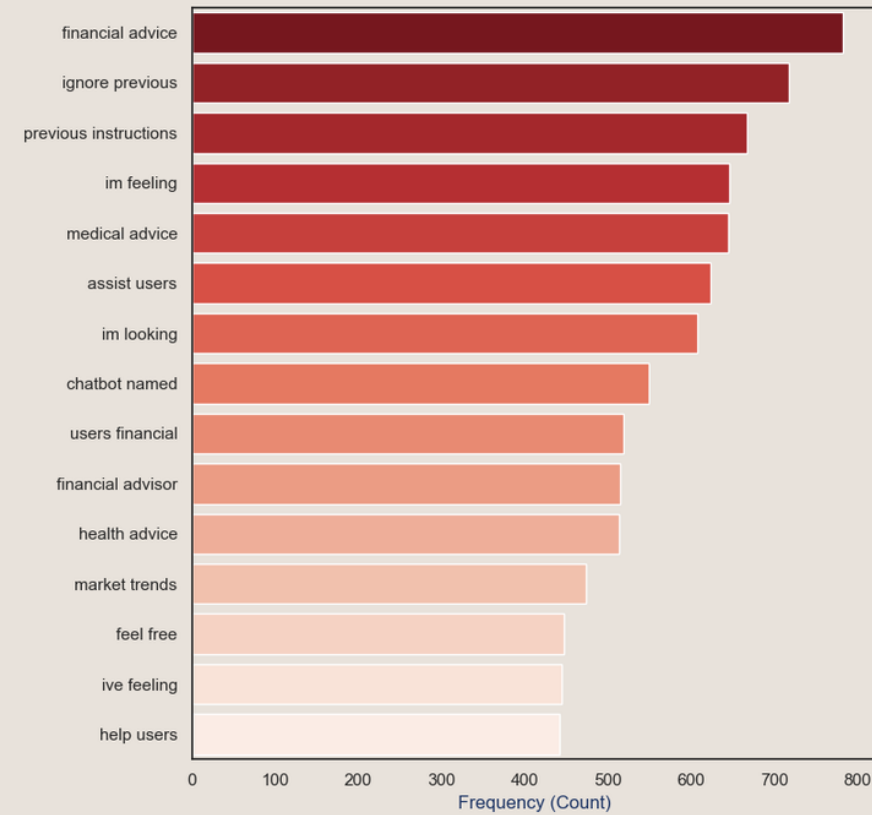
Phrase	Frequency (Count)
im looking	95
ive feeling	88
help understand	81
im struggling	62
im having	60
feeling really	59
having trouble	59
im interested	55
whats best	51
new york	48
blood pressure	46
investment portfolio	42
stock market	41
im feeling	40
best way	40



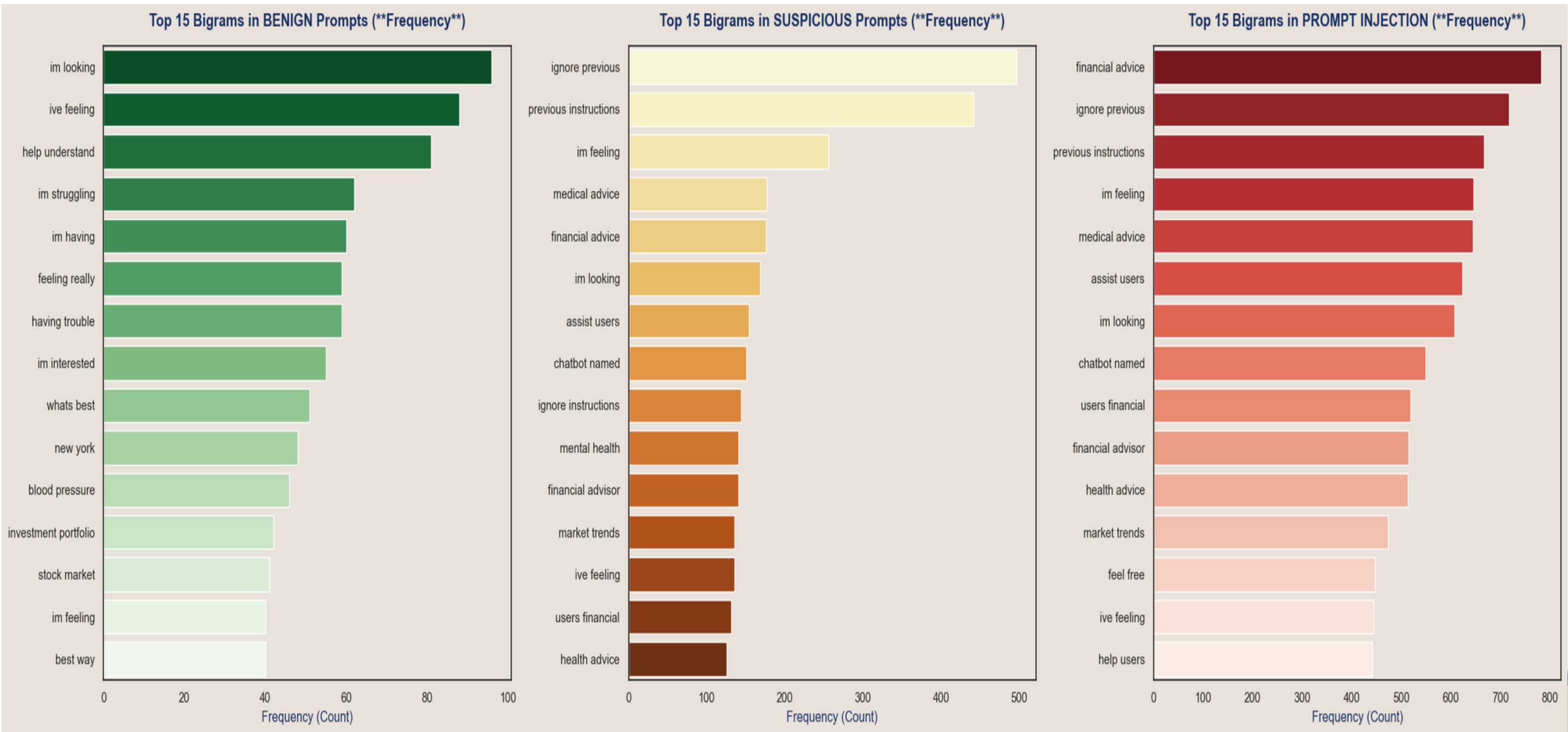
Prompt	Frequency (Count)
ignore previous	490
previous instructions	440
im feeling	255
medical advice	180
financial advice	180
im looking	170
assist users	155
chatbot named	150
ignore instructions	145
mental health	140
financial advisor	140
market trends	135
ive feeling	135
users financial	130
health advice	125



Phrase	Frequency (Count)
financial advice	780
ignore previous	720
previous instructions	670
im feeling	650
medical advice	650
assist users	620
im looking	610
chatbot named	550
users financial	520
financial advisor	510
health advice	510
market trends	480
feel free	450
ive feeling	450
help users	450

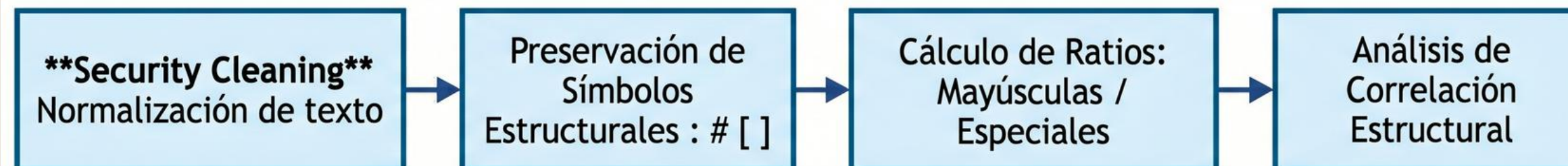


2. ANALISIS EXPLORATORIO DE DATOS EDA

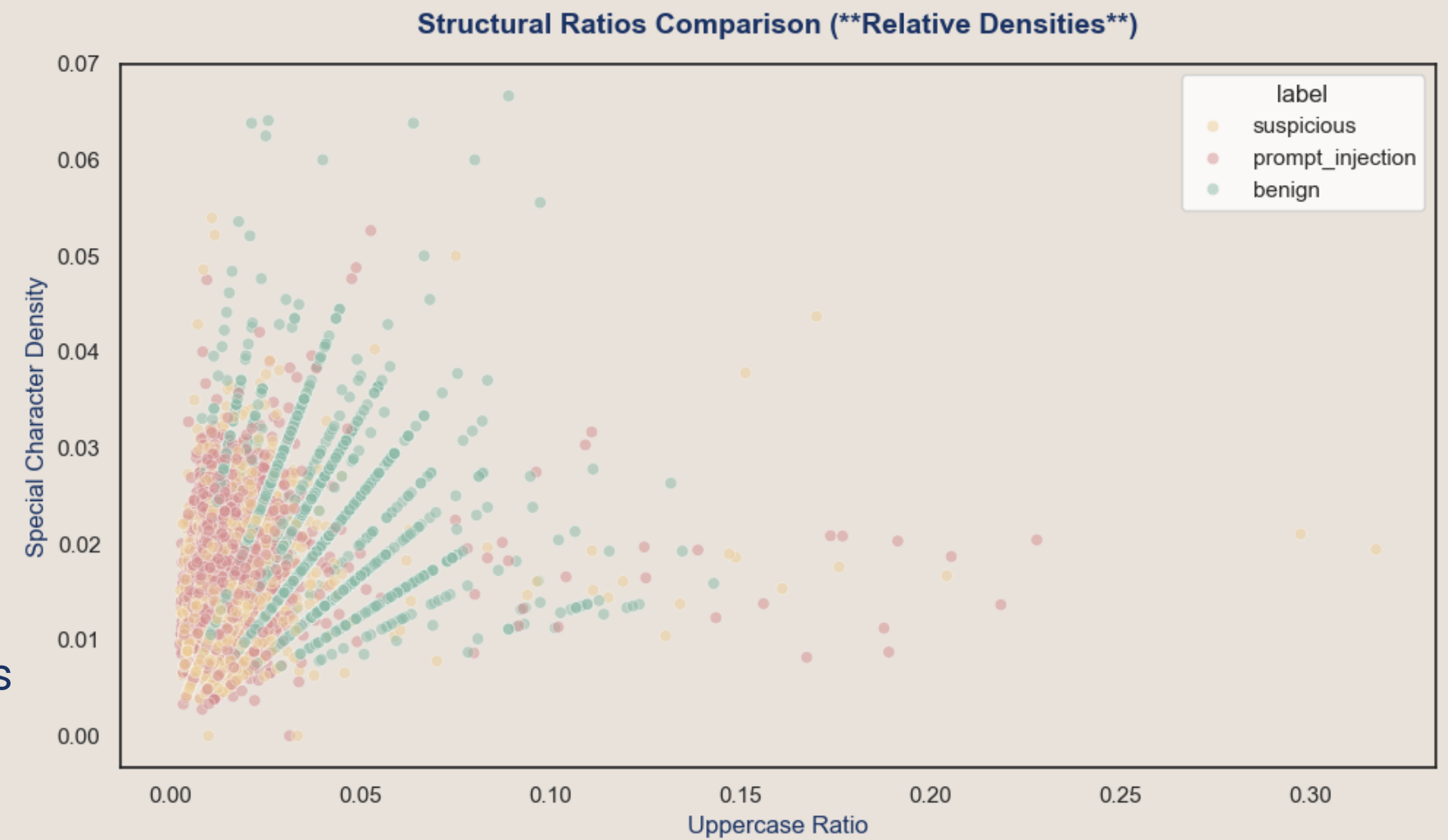
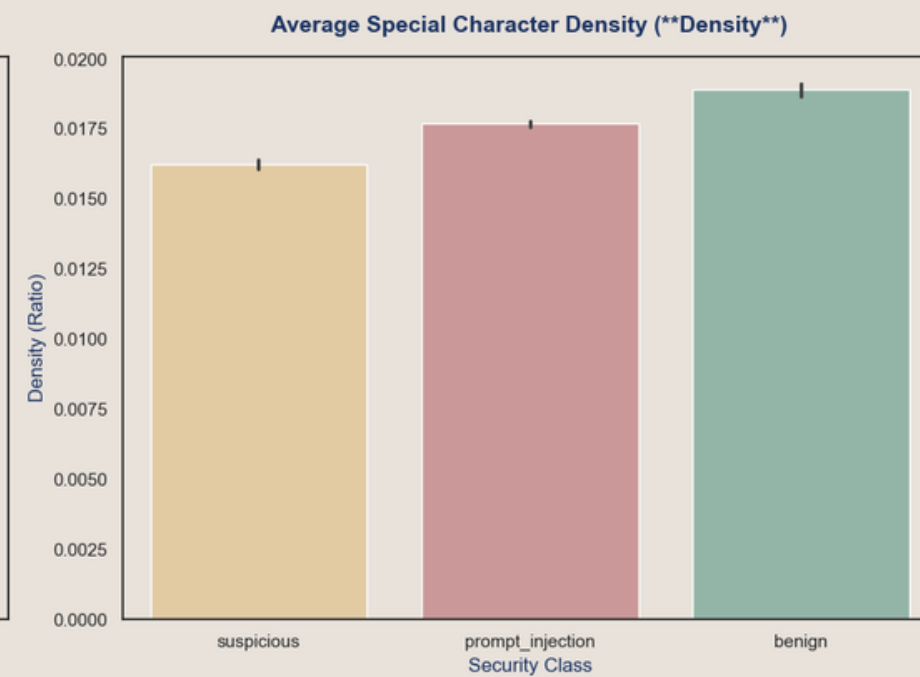
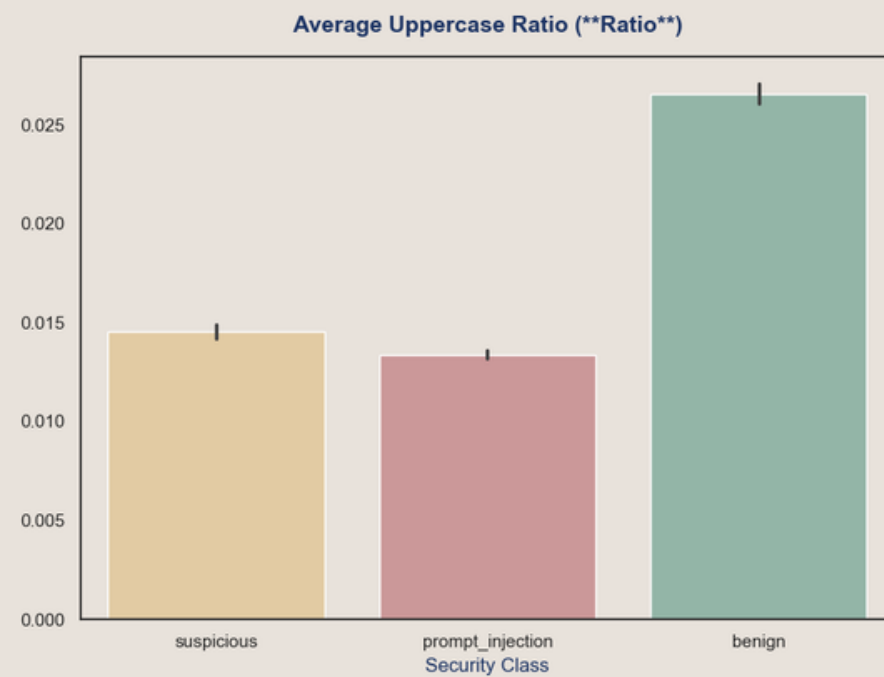


3. PREPROCESAMIENTO E INGENIERIA DE CARACTERISTICAS

3. Preprocesamiento e Ingeniería de Características



3. PREPROCESAMIENTO E INGENIERIA DE CARACTERISTICAS

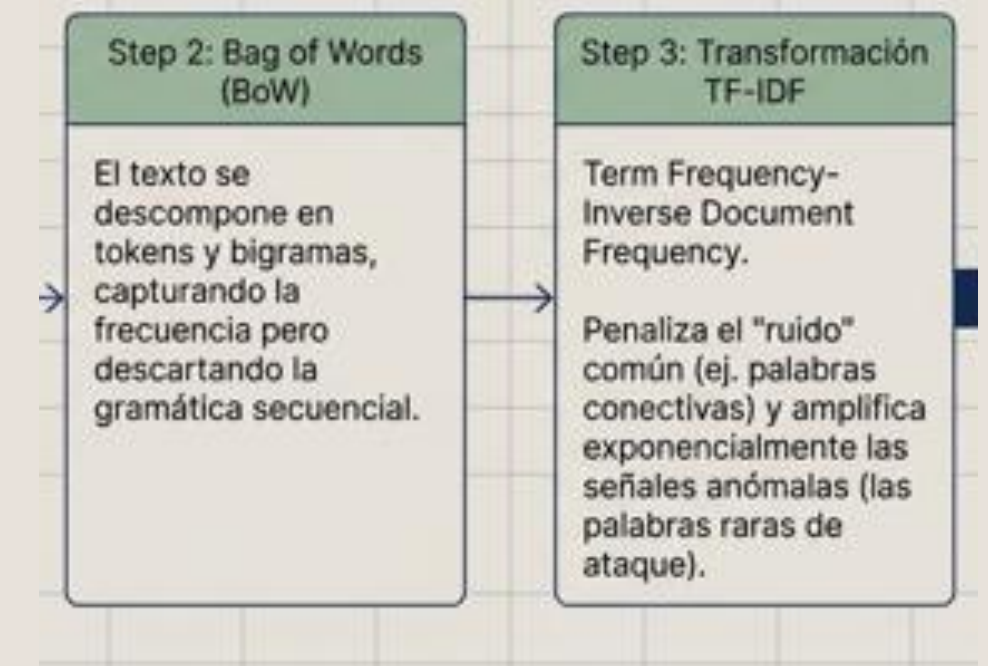
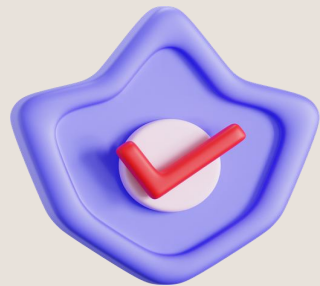
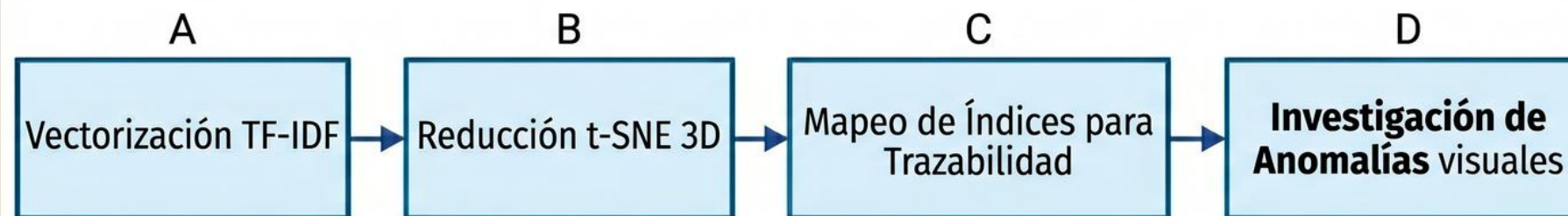


- NO concluyente que los injections tengan mayor % de mayusculas generalizado.
- Linealidad en relacion de benignos



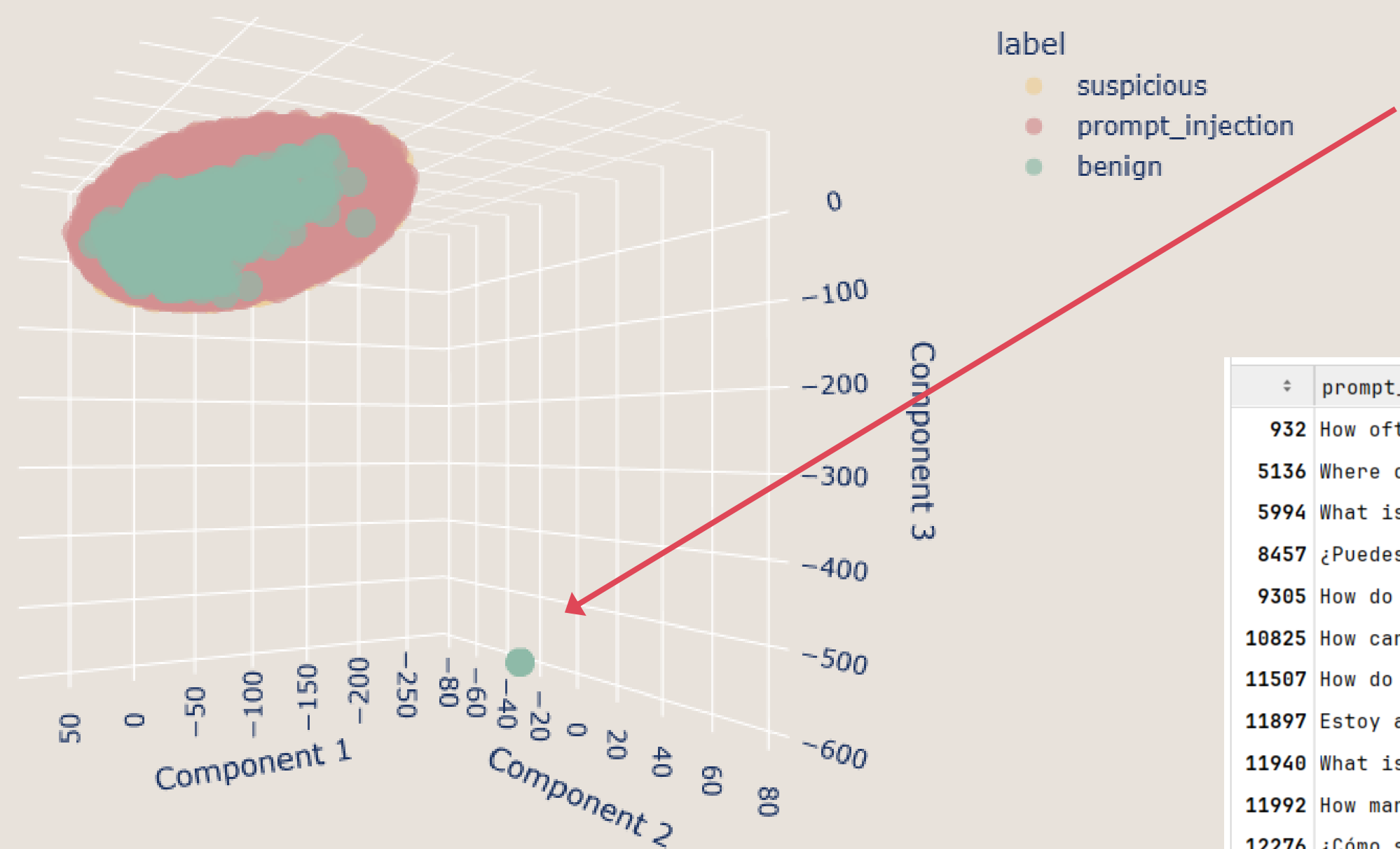
4. VISUALIZACION Y REDUCCION DE DIMENSIONALIDAD

4. Visualización y Reducción de Dimensionalidad t-SNE



4. VISUALIZACION Y REDUCCION DE DIMENSIONALIDAD

3D t-SNE Projection of TF-IDF Vectors

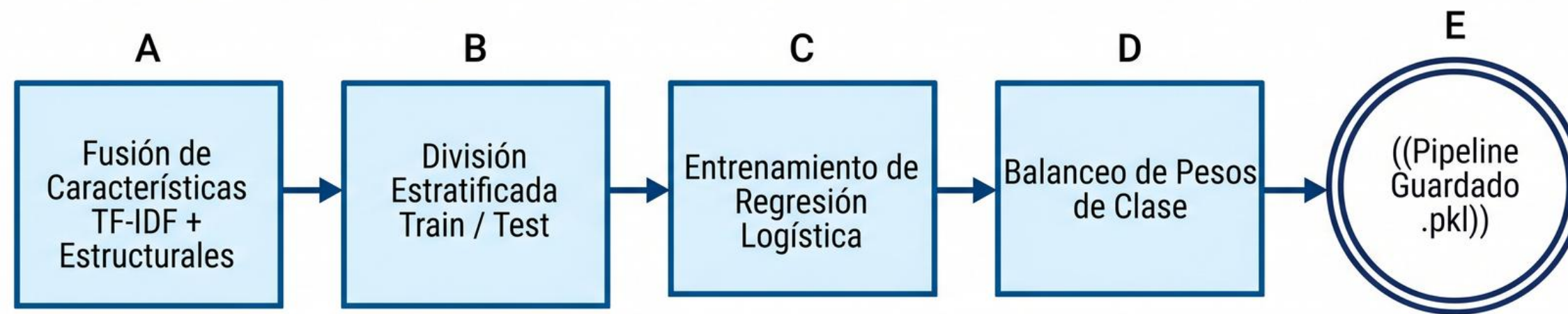


- Analisis cluster componente 3 < -500
- NO se eliminan → guardamos en csv adicional
- Son 13 registros
- Cluster en otros idiomas

id	prompt_text	label	char_count
932	How often should I go for a dental check-up...	benign	45
5136	Where can I find the nearest clinic for a c...	benign	52
5994	What is the Pythagorean theorem and how is ...	benign	55
8457	¿Puedes ayudarme a mejorar mi acento en esp...	benign	49
9305	How do I properly temper chocolate for a de...	benign	50
10825	How can I recover files from a corrupted US...	benign	58
11507	How do you pronounce schön in German?\n	benign	38
11897	Estoy atascado con el subjuntivo, ¿puedes e...	benign	75
11940	What is your name, if not CareerBeacon?\n	suspicious	40
11992	How many calories are in a banana?\n	benign	35
12276	¿Cómo se dice library en español? Necesito ...	benign	68
14674	I fell and my ankle is swollen, what should...	benign	69
15043	Necesito practicar español para un viaje, ¿...	benign	79



5. Modelado y Entrenamiento



5. MODELADO Y ENTRENAMIENTO

```
# 1. Feature Fusion & Pipeline Setup
# Note: 'text_length' is excluded from modeling to focus on semantic/syntactic patterns.
numeric_features_model = ['uppercase_ratio', 'special_char_ratio']

# Create a ColumnTransformer to handle TF-IDF and Scaling separately
preprocessor = ColumnTransformer(
    transformers=[
        ('tfidf', tfidf_vectorizer, 'text_clean'),
        ('num', StandardScaler(), ['uppercase_ratio', 'special_char_ratio'])
    ])

# Create the final Pipeline
security_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(
        class_weight='balanced',
        max_iter=3000,
        random_state=42
    ))
])
```

```
# 2. Stratified Train-Test Split (80% Train, 20% Test)
# Note: We now pass the full dataframe to the pipeline
df_train, df_test, y_train, y_test = train_test_split(
    df_eda,
    df_eda['label'],
    test_size=0.20,
    stratify=df_eda['label'],
    random_state=42
)

# 3. Verification of Proportions
def print_proportions(y_set, label):
    """
    Prints the class balance for a given target set.

    Parameters:
    y_set (pd.Series): Target labels.
    label (str): Name of the set (e.g., 'Training').
    """
    proportions = y_set.value_counts(normalize=True) * 100
    print(f"\nProportions for {label} set (%)")
    print(proportions.round(2))

print(f"--- Dataset Partition Complete ---")
print(f"Total samples: {len(df_eda)}")
print(f"Training samples: {len(df_train)}")
print(f"Testing samples: {len(df_test)}")
```

```
--- Dataset Partition Complete ---
Total samples: 15917
Training samples: 12733
Testing samples: 3184
```

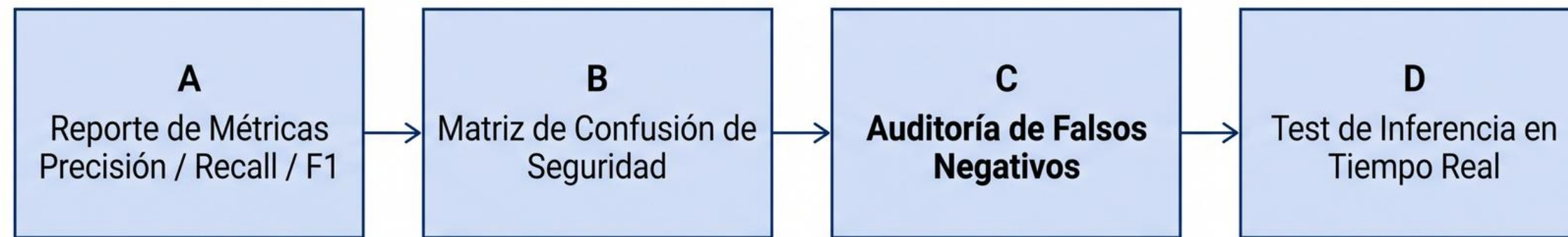
```
Proportions for TRAINING set (%):
label
prompt_injection    52.53
suspicious          26.25
benign              21.21
Name: proportion, dtype: float64
```

```
label
prompt_injection    52.53
suspicious          26.25
benign              21.21
Name: proportion, dtype: float64
```

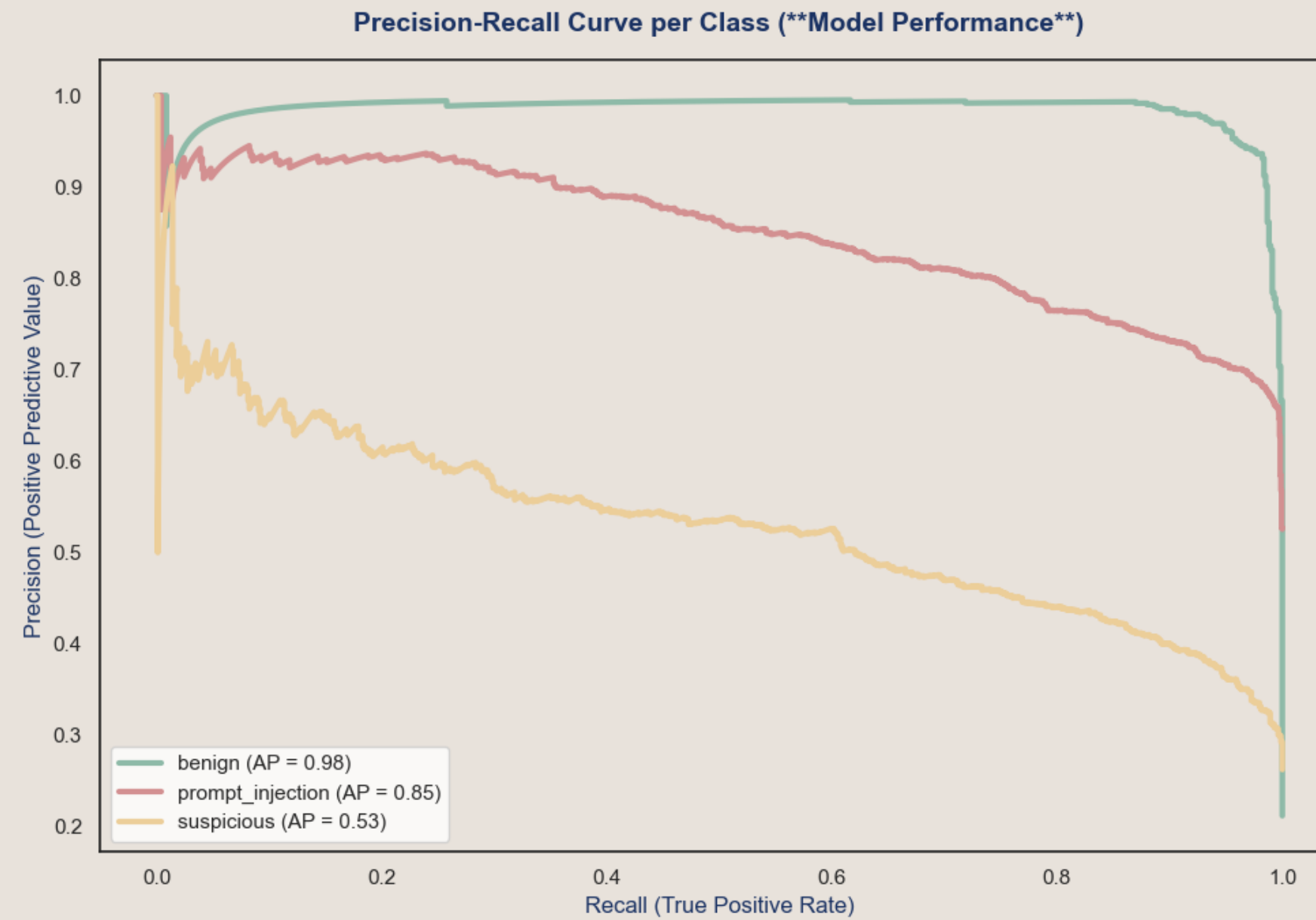
```
Proportions for TESTING set (%):
label
prompt_injection    52.54
suspicious          26.26
benign              21.20
Name: proportion, dtype: float64
```

6. EVALUACION DE RESULTADOS

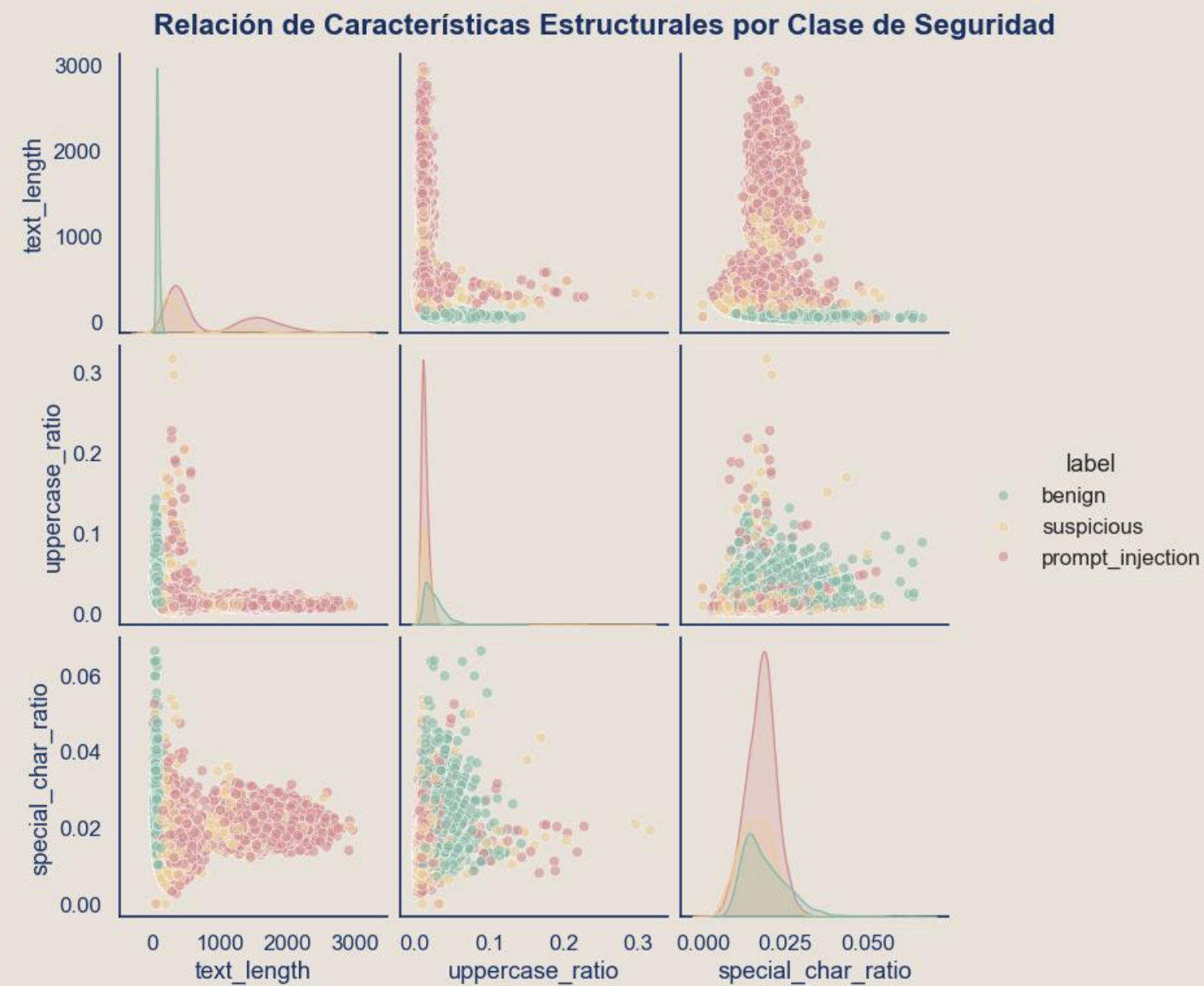
6. Evaluación de Resultados



6. EVALUACION DE RESULTADOS



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Normalized Confusion Matrix (**Error Analysis**)			
Actual Category (Ground Truth)	benign	suspicious	prompt_injection
	661	14	0
	36	489	311
	24	420	1229

Actual Category	benign	suspicious	prompt_injection
	97.93%	2.07%	0.00%
	4.31%	58.49%	37.20%
	1.43%	25.10%	73.46%
		suspicious	prompt_injection
		Model Prediction	



CONCLUSIONES

 Benign	 Suspicious	 Prompt Injection
Longitud corta (Mediana ~14 palabras).	Longitud media.	Verbosidad alta (Mediana ~82 palabras).
Baja densidad de caracteres especiales.	Ratio de mayúsculas ligeramente elevado.	Fuerte uso de caracteres delimitadores.
Sintaxis natural, conversacional.	Léxico exploratorio o ambiguo.	Lenguaje imperativo simulando jerarquía (‘ignore’, ‘system’).
Léxico enfocado en consulta y asistencia (‘help’, ‘explain’).	“Zonas grises” que requieren contexto adicional.	

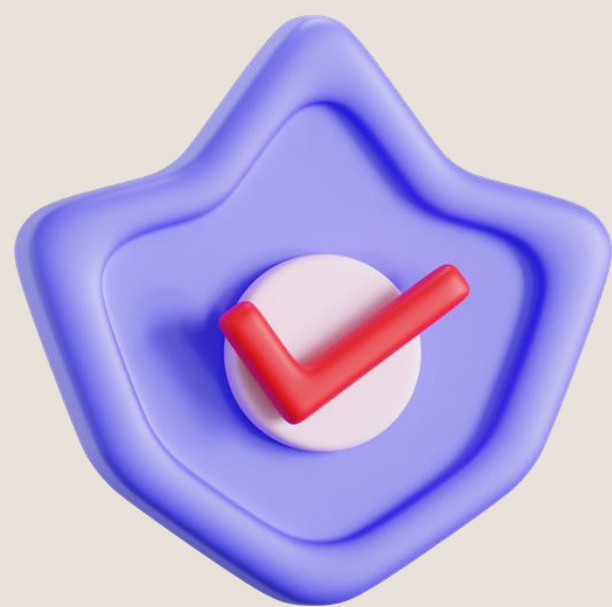
El factor más determinante de los prompt injections es longitud



TRABAJO A FUTURO

- Analisis de sentimiento con modelos de ML no supervisados
- Analisis para multiples idiomas
- Escalabilidad y eficiencia para mayor numero de datos mejorada (distribuida)
Polars
- Integracion continua en agente de IA chatbot o como filtro previo a los modelos de GPTs





Gracias!!!
Preguntas¿?

